INCH-POUND

MIL-R-874
AMENDMENT 3
10 February 1994
SUPERSEDING
AMENDMENT 2
28 April 1993

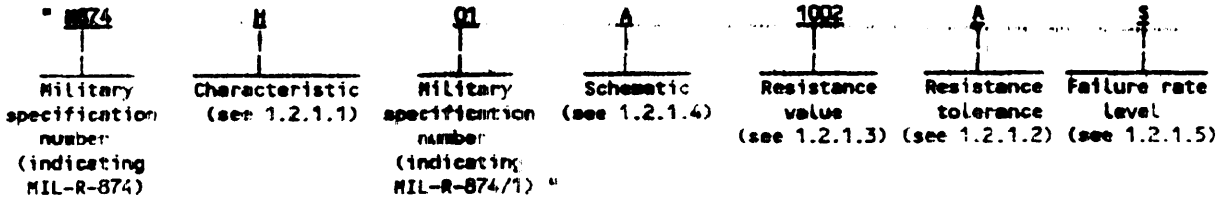
MILITARY SPECIFICATION

RESISTOR NETWORKS, FIXED, FILM,
ESTABLISHED RELIABILITY,
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-874, dated 21 June 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1, delete and substitute the following PIN number:




PAGE 7

3.2, delete and substitute the following:

"3.2 Qualification. Networks furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time of award of contract (see 4.4 and 6.3). Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL."

PAGE 8

Add the following new paragraphs:

"3.5.2.1 Solder dip (retinning) leads. The manufacturer (or authorized category B or C distributor) may solder dip/retin the leads of product supplied to this specification, provided the solder dip process has been approved by the qualifying activity.

"3.5.2.2 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276 (NOTE: The 200 microinch maximum thickness is not applicable). The manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the product.


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b. When the lead originally qualified was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in 3.5.2.2a, approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder dip process, the resistors are subjected to the dc resistance test (and other group A electricals). No defects are allowed.

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to solder heat test, followed by the moisture resistance test. (NOTE: Solder dip of gold-plated leads is not allowed.)"

"3.5.2.3 Solder dip/retinning options. The manufacturer (or authorized category B or C distributor) may solder dip/retin as follows:

a. After the 100 percent group A screening tests. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening tests shall be repeated on 100 percent of the lot. (NOTE: The manufacturer may solder dip/retin prior to the 100 percent electrical measurements of the group A, subgroup 1 tests.) The percentage defective allowable (PDA) for the electrical measurements shall be as for the subgroup 1 tests.

b. As a corrective action if the lot fails the group A solderability test.

c. After the group A inspection has been completed. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening test shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Following these tests, the manufacturer shall submit the lot to the group A solderability test as specified in 4.8.6."

Add the following new paragraph:

"3.5.5 Bonding. If thermocompression bonding is used for internal connections, a gold to gold bond shall form a solid phase weld (see 3.1)."

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Add the following new paragraph:

"3.5.6 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat effective 6 months from the date of this specification (see 6.4.3.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

PAGE 15

TABLE VII, delete and substitute:

" TABLE VII. Failure rate substitution.

| Failure rate | Failure rate substitution |
|---|---|
| S | — |
| R | S |
| P | S, R |
| M | S, R, P |

"



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PART OR IDENTIFYING NUMBER (PIN): M39010/01- (plus dash number from table I and suffix

TABLE VIII, delete and substitute:

" TABLE VIII. Substitution of temperature characteristics.

| Characteristic | Characteristic substitution |
|---|---|
| H | --- |
| K | H |
| M | H, K |

"

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TABLE IX, delete and substitute:

" TABLE IX. Resistance tolerance substitution.

| Resistance tolerance | Resistance tolerance substitution |
|---|---|
| B | --- |
| D | B |
| F | B, D |
| G | B, D, F |
| J | B, D, F, G |

"



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* TABLE X, delete and substitute:

* TABLE X. Qualification inspection.

| Inspection | Number of sample units | Requirement paragraph | Method paragraph | Number of 1/ failures allowed |
|---|---|---|---|---|
| Group I 2/ 3/ | | | | |
| Thermal shock | All sample units 4/ | 3.8 | 4.8.3 | 5/ |
| Power conditioning | | 3.9 | 4.8.4 | |
| DC resistance | | 3.10 | 4.8.5 | 5/ |
| Group IA 3/ 6/ | | | | |
| Visual and mechanical inspection | | 3.1, 3.4, 3.5 3.28, 3.29 | 4.8.2 | 0 |
| Group IB 7/ | | | | |
| Solderability | | 3.11 | 4.8.6 | 1 |
| Group II 7/ | | | | |
| Resistance to solvents | 12 4/ | 3.12 | 4.8.7 | 1 |
| Group III | | | | |
| Resistance-temperature characteristics | 20 or 30 | 3.13 | 4.8.8 | |
| Low temperature operation | 10 high | 3.14 | 4.8.9 | |
| Short-time overload | 10 critical 8/ | 3.15 | 4.8.10 | |
| Terminal strength | 10 low | 3.16 | 4.8.11 | 1 9/ |
| Group IV | | | | |
| Thermal shock | | 3.8 | 4.8.3 | |
| Dielectric withstanding voltage | 20 or 30 | 3.17 | 4.8.12 | |
| Insulation resistance | 10 high | 3.18 | 4.8.13 | |
| Resistance to soldering heat 10/ | 10 critical 8/ | 3.19 | 4.8.14 | |
| Moisture resistance | 10 low | 3.20 | 4.8.15 | 1 9/ |
| Group V | | | | |
| Shock (specified pulse) | 20 or 30 | | | |
| Vibration, high frequency | 10 high | 3.21 | 4.8.16 | |
| | 10 critical 8/ | 3.22 | 4.8.17 | |
| | 10 low | | | 1 9/ |

See footnotes at end of table.



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* TABLE X. Qualification inspection - Continued.

| Inspection | Number of sample units | Requirement paragraph | Method paragraph | Number of failures allowed |
|---|---|---|---|---|
| Group VI<br>Life | 102<br>34 high<br>34 critical<br>34 low | 3.23 | 4.8.18 | 1 9/ |
| Group VIA<br>Power rating at 25°C | 20 or 30<br>10 high<br>10 critical 8/<br>10 low | 3.23.1 | 4.8.18 | 1 9/ |
| Group VII<br>High temperature exposure<br>Low temperature storage | 20 or 30<br>10 high<br>10 critical 8/<br>10 low | 3.24<br>3.25 | 4.8.19<br>4.8.20 | 1 9/ |
| Group VIII<br>Fungus 4/ 11/ | 10 | 3.26 | 4.8.21 | 0 |

1/ Failure of a single network in one or more tests of a group shall be charged as a single defective.
2/ Tests need not be performed if group A inspection has been performed on the qualification samples.
3/ Nondestructive tests.
4/ Sample units for group II and VIII are not required to be subjected to group I.
5/ Networks shall meet all requirements of group I before subjecting to groups III through VII.
6/ Marking shall be considered defective if the marking is illegible or incorrect.
7/ Tests may be performed on electrical rejects.
8/ When no critical value is specified, only the highest and lowest resistance values shall be tested (20 samples, total).
9/ Number of defective packages allowed per resistance value tested (e.g., low, critical, and high resistance).
10/ The internal visual examination shall be performed after the final inspection of group IV.
11/ The fungus requirement is either by certification or performance."

PAGE 19

4.4.4.1c, delete and substitute the following:

"c. Procedure III - Maintenance of FR levels qualification. Maintenance period B of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.7.2.1). For FR levels R and S data from all lead types may be combined."

4.5.1, delete in its entirety.

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4.7.1.2.3, last sentence: Delete in its entirety.

After 4.7.1.2.3 and before table XI, add the following new paragraphs:

"4.7.1.2.4 Subgroup 4 (solderability).

* "4.7.1.2.4.1 Sampling plan. Five samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. If there are one or more defects, the lot shall be considered to have failed.



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"4.7.1.2.4.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.8.6. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in b. "

4.7.1.2.4.2b delete and substitute the following paragraph (alters a change from amendment 1):

"b. The manufacturer submits the failed lot to a 100-percent solder dip using an approved solder dip process in accordance with 3.5.2.1. Following the solder dip, the electrical measurements required in group A, subgroup 1 tests shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Thirteen additional samples shall then be selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of this specification."

"4.7.1.2.4.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract."

* TABLE XI, delete and substitute the following table:

"TABLE XI. Group A inspection.

| Inspection | Requirement paragraph | Test method paragraph | Sampling procedure |
|---|---|---|---|
| Subgroup 1 | | | |
| Thermal Shock | 3.8 | 4.8.3 | |
| Power conditioning | 3.9 | 4.8.4 | 100 percent |
| DC resistance 1/ | 3.10 | 4.8.5 | inspection |
| Subgroup 2 (ppa) | | | |
| DC resistance (ppa-2) | 3.10 | 4.8.5 | See table |
| Mechanical examination (ppa-3) | 3.5 | 4.8.2 | XII |
| Subgroup 3 | | | |
| Visual examination | 3.1, 3.4, 3.27 | 4.8.2 | See 4.7.1.2.3 |
| Subgroup 4 | | | |
| Solderability 2/ | 3.11 | 4.8.6 | See 4.7.1.2.4 |

1/ Networks shall meet the specified initial resistance tolerance. The resistance measurement made upon completion of the power conditioning test may be used if a measurement has been made which can, without conversion, be directly related to nominal resistance value and tolerance.

2/ As an option, solderability test may be performed on electrical rejects or empty body networks. The empty body networks must be processed through all the standard manufacturing and test flow cycles, oven bakes, and temperature cycles of qualified networks to assure that the networks leads and printed surfaces are in the usual condition prior to test.



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* TABLE XII, delete and substitute the following table: (alters change in amendment 2)

" TABLE XII. Sampling plans for ppm categories.

| Lot size | Sample size PPM-2 | Sample size PPM-3 |
|---|---|---|
| 1 - 13 | 100% | 100% |
| 14 - 125 | 100% | 13 |
| 126 - 150 | 125 | 13 |
| 151 - 280 | 125 | 20 |
| 281 - 500 | 125 | 29 |
| 501 - 1,200 | 125 | 34 |
| 1,201 - 3,200 | 125 | 42 |
| 3,201 - 10,000 | 125 | 50 |
| 10,001 - 35,000 | 294 | 60 |
| 35,001 - 150,000 | 294 | 74 |
| 150,001 - 500,000 | 345 | 90 |
| 500,001 and over | 435 | 102 |

"

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TABLE XIII, delete in its entirety.

* 4.7.1.3, delete and substitute: (alters change from amendment 2).

"4.7.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table XIV, in the order shown. The tests shall be performed on sample units from lots that have been subjected to and have passed the group A inspection."

* TABLE XIV, delete and substitute the following table: (alters change in amendment 2).

"TABLE XIV. Group B inspection.

| Inspection | Requirement paragraph | Test method paragraph | Number of samples | Number of failures allowed |
|---|---|---|---|---|
| Subgroup 1 1/ | | | | |
| Resistance temperature characteristic | 3.13 | 4.8.8 | 13 | 0 |
| Subgroup 2 1/ 2/ | | | | |
| Resistance to solvents | 3.12 | 4.8.7 | 8 | 0 |

1/ If the manufacturer can demonstrate that this test has been performed for five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.
2/ Tests may be performed on electrical rejects. "

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* 4.7.2.1.1.2, delete and substitute the following:

"4.7.2.1.1.2 Subgroup 2. Ten sample units with resistance values representative of production over that period shall be inspected every month. One defective unit shall be allowed."



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* 4.7.2.1.2, delete and substitute the following:

"4.7.2.1.2 Quarterly. Ten sample units with resistance values representative of production over that period shall be inspected quarterly. Ten sample units from the resistance decade produced shall be subjected to these tests. One defective unit shall be allowed. "

* 4.7.2.1.3, delete and substitute the following:

"4.7.2.1.3 Semiannually. Ten sample units with resistance values representative of that production over that period shall be subjected to the tests of subgroup 1. Ten sample units with resistance values representative of production over that period shall be subjected to the tests of subgroup 2. One defective unit shall be allowed. "

* TABLE XV, delete and substitute the following table:

TABLE XV. Group C inspection.

| Inspection | Requirement paragraph | Test method paragraph | Number of sample units for inspection | Number of failures allowed |
|---|---|---|---|---|
| Monthly<br>Subgroup 1<br>Life | 3.23 | 4.8.18 | 4.7.2.1.1 | 4.7.2.1.1.1 |
| Subgroup 2 1/<br>Thermal shock<br>Dielectric withstanding voltage<br>Insulation resistance<br>Low temperature operation<br>Short time overload<br>Terminal strength | 3.8<br>3.17<br>3.18<br>3.14<br>3.15<br>3.16 | 4.8.3<br>4.8.12<br>4.8.13<br>4.8.9<br>4.8.10<br>4.8.11 | 10 | 1 |
| Quarterly<br>Resistance to soldering heat 2/<br>Moisture resistance | 3.19<br>3.20 | 4.8.14<br>4.8.15 | 10 | 1 |
| Semiannually<br>Subgroup 1 1/<br>Shock (specified)<br>Vibration (high frequency) | 3.21<br>3.22 | 4.8.16<br>4.8.17 | 10 | 1 |
| Subgroup 2 3/<br>High temperature exposure<br>Low temperature storage | 3.24<br>3.25 | 4.8.19<br>4.8.20 | 10 | |

1/ If the manufacturer can demonstrate that these tests have been performed for five consecutive times with zero failures, these tests, with the approval of the quality activity, can be deleted. The manufacturer, however, shall perform these tests every 3 years after the deletion as part of long term design verification. If the design, material, construction or processing of the part is changed, or if there are any problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute.
2/ The internal inspection shall be performed after the final inspection of the subgroup.
3/ If the manufacturer can demonstrate that this test has been performed for five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency. "



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4.8.5a, delete and substitute the following paragraph:

"a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent or better."

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TABLE XVII, delete and substitute:

" TABLE XVII. Resistance-temperature characteristic.

| Sequence | Temperature | |
|---|---|---|
| | Qualification inspection | Group B acceptance inspection 1/ |
| | °C | °C |
| 1 | Room temperature 2/ | Room temperature 2/ |
| 2 | -15 ±3 | -55 ±3 |
| 3 | -55 ±3 | Room temperature 2/ |
| 4 | Room temperature 2/ | 125 ±3 |
| 5 | 65 ±3 | --- |
| 6 | 125 ±3 | --- |

1/ At the option of the manufacturer, the reverse sequence may be as follows:

| | |
|---|---|
| 1 | Room temperature 1/ |
| 2 | +125 ±3 |
| 3 | Room temperature 1/ |
| 4 | -55 ±3 |

2/ This temperature shall be considered the reference temperature for each of the succeeding temperatures. "

PAGE 35

Add the following new paragraph:

"6.4.3.1 Tin plating finishes. Tin plating is prohibited (see 3.5.6) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter, refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin)."

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30.3, delete and substitute:

"30.3 Description of items. The manufacturer shall submit a detailed description of the network being submitted for inspection, including materials used for the resistance element and protective enclosure or coating. After qualification has been granted, no changes shall be made in materials, design, or construction without proper notification to the qualifying activity."



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40.1, delete and substitute:

"40.1 Extension of qualification. The resistance range included in the qualification of any one network style will be between the lowest and highest resistance values which pass the qualification inspection (see 3.1). Qualification of one characteristic is basis for qualification of another characteristic, as indicated in table XIX. Qualification of the lower resistance tolerances will qualify for the higher resistance tolerances in accordance with table XX. Extent of qualification by style shall be in accordance with table XXI. Internal welded construction will qualify a nonwelded construction. Extent of qualification by schematic shall be in accordance with table XXII. As a requisite for extension of qualification as described herein between characteristics, tolerances, schematics, resistance values and failure rates, the product involved must be manufactured using the same facilities, processes, and materials as the product originally submitted for qualification."

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TABLE XVIII, delete and substitute:

" TABLE XVIII. Critical resistance value for qualification inspection. 1/

| Style | Critical resistance value 1/ 2/ | | | | | |
|---|---|---|---|---|---|---|
| | Schematic A | Schematic B | Schematic C | Schematic G | Schematic J | Schematic H |
| | Characteristics H, K, M | Characteristics H, K, M | Characteristics H, K, M | Characteristics H, K, M | Characteristics H, K, M | Characteristics H, K, M |
| | MΩ | MΩ | MΩ | MΩ | MΩ | MΩ |
| RZR10 | 0.047 | 0.1 | | | .200 | |
| RZR11 | 0.047 | 0.1 | | | .200 | |
| RZR20 | 0.047 | 0.1 | | | .200 | |
| RZR21 | 0.047 | 0.1 | | | .200 | |
| RZR30 | 0.047 | 0.1 | | | .160 | |
| RZR31 | 0.047 | 0.1 | | | .160 | |
| RZR40 | | | .012 | .012 | | .020 |
| RZR41 | | | .012 | .012 | | .020 |
| RZR50 | | | .012 | .012 | | .020 |
| RZR51 | | | .012 | .012 | | .020 |
| RZR60 | | | .012 | .012 | | .020 |
| RZR61 | | | .012 | .012 | | .020 |
| RZR70 | | | .020 | .020 | | .033 |
| RZR71 | | | .020 | .020 | | .033 |
| RZR80 | | | .020 | .020 | | .033 |
| RZR81 | | | .020 | .020 | | .033 |
| RZR90 | | | .020 | .020 | | .033 |
| RZR91 | | | .020 | .020 | | .033 |
| RZR95 | 0.047 | 0.10 | | | .160 | |
| RZR96 | 0.047 | 0.10 | | | .160 | |

1/ Maximum continuous working voltage shall be applied (see 3.1).
2/ The critical resistance value is the maximum standard resistance value which will dissipate full wattage when maximum continuous working voltage is applied. "



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TABLE XXI, delete and substitute:

" TABLE XXI. Extent of qualification by style.

| Style | Will qualify style(s) |
|---|---|
| RZR10 | RZR10 |
| RZR11 | RZR11, RZR10 |
| RZR20 | RZR20, RZR10 |
| RZR21 | RZR21, RZR20, RZR11, RZR10 |
| RZR30 | RZR30 |
| RZR31 | RZR31, RZR30 |
| RZR40 | RZR40 |
| RZR41 | RZR41, RZR40 |
| RZR50 | RZR50, RZR40 |
| RZR51 | RZR51, RZR50, RZR41, RZR40 |
| RZR60 | RZR60, RZR50, RZR40 |
| RZR61 | RZR61, RZR60, RZR51, RZR50, RZR41, RZR40 |
| RZR70 | RZR70 |
| RZR71 | RZR71, RZR70 |
| RZR80 | RZR80, RZR70 |
| RZR81 | RZR81, RZR80, RZR71, RZR70 |
| RZR90 | RZR90, RZR80, RZR70 |
| RZR91 | RZR91, RZR90, RZR81, RZR80, RZR71, RZR70 |
| RZR95 | RZR95, RZR30 |
| RZR96 | RZR96, RZR95, RZR31, RZR30 |

"

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, CG, MC, OS
Air Force - 17, 19
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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